

Qatar-9b

R-2239

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-9_260429 · created 2026-08-15T01:59:53+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461159.873 +/- 0.014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.205 +/- 0.059
Transit depth (Rp/Rs)^2	4.19 +/- 2.43 [%]
Orbital Inclination (inc)	88.79 +/- 1.67
Airmass coefficient 1 (a1)	0.43 +/- 0.25
Airmass coefficient 2 (a2)	0.43 +/- 0.37
Scatter in the residuals of the lightcurve fit is	45.62 %
Best Comparison Star	#3 - [593, 373]
Optimal Aperture	0.93
Optimal Annulus	6.66
Transit Duration (day)	0.072 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.205 ± 0.059 vs 0.1489 (deviation 0.58σ)
Duration fit vs published	0.072 d vs 0.0779 d (deviation 0.36σ)
Mid-time O–C	29.9 min
Depth SNR	2.1
Residual scatter	45.62 %

Light curve

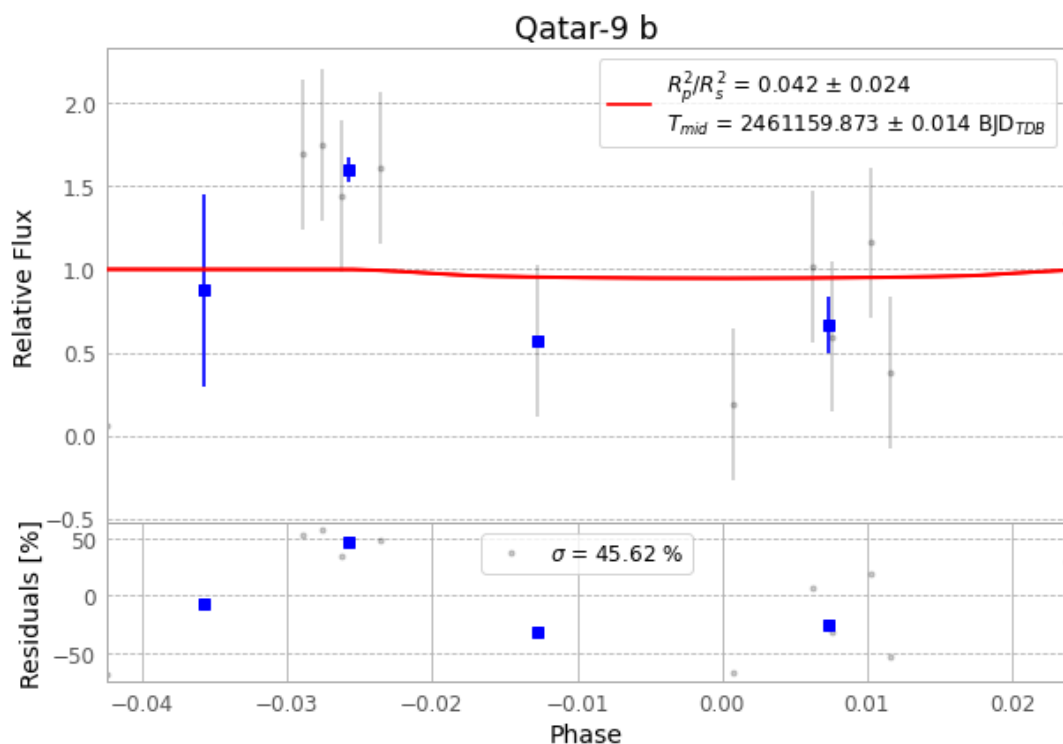


Figure 1 — FinalLightCurve_Qatar-9 b_2026-04-29.png (SHA-256 5325f72636e562d5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [451, 78], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 47f4401e861359c7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

61 FITS frames (40 MB), Qatar-9260429072115.FITS → Qatar-9260429102116.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-9-260429.log (6636f268d401...), FinalLightCurve_Qatar-9 b_2026-04-29.png (5325f72636e5...), FinalLightCurve_Qatar-9 b_2026-04-29.csv (68cf14b80663...)

Compute resources

Wall time 118.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 01:59Z.